

Chapter 10: System Engineering

System Engineering

Software engineering occurs as a consequence of a process called system engineering. Instead of concentrating solely on software, system engineering focuses on a variety of elements, analyzing, designing, and organizing those elements into a system that can be a product, a service, or a technology for the transformation of information or control.

Business Process Engineering

- Focus on a business enterprise

Product Engineering

- Focus on a product to be built

What is a “System”?

Types of systems:

- political, educational, avionics, banking, manufacturing, ...

Computer-based system:

“A set of elements that are organized to accomplish some predefined goal by processing information”

Goals: Support a business function, develop a product that can be sold to generate business revenue. To accomplish the goal, a computer-based system makes use of a variety of system elements.

System Elements

- **Software.** Computer programs, data structures, and related documentation that serve to effect the logical method, procedure, or control that is required.
- **Hardware.** Electronic devices that provide computing capability, the interconnectivity devices that enable the flow of data, and electromechanical devices that provide external world function.
- **People.** Users and operators of hardware and software.
- **Database.** A large, organized collection of information that is accessed via software.
- **Documentation.** Descriptive information that portrays the use and/or operation of the system.
- **Procedures.** The steps that define the specific use of each system element or the procedural context in which the system resides.

System Engineering Hierarchy

Each computer-based system can be part of a larger system

System Engineering Hierarchy

- Organize the systems into a set of layered views
- Define the elements for a specific computer-based system in the context of the overall hierarchy of systems

Information
Strategy Planning

System Engineering Hierarchy

Domain of interest

Business or
product domain

World view

Business Area
Analysis

System element

Domain view

Element view

Software
Engineering

Detailed view



Stated in a slightly more formal manner, the world view (WV) is composed of a set of domains (D_i), which can each be a system or system of systems in its own right.

$$WV = \{D_1, D_2, D_3, \dots, D_n\}$$

Each domain is composed of specific elements (E_j) each of which serves some role in accomplishing the objective and goals of the domain or component:

$$D_i = \{E_1, E_2, E_3, \dots, E_m\}$$

Finally, each element is implemented by specifying the technical components (C_k) that achieve the necessary function for an element:

$$E_j = \{C_1, C_2, C_3, \dots, C_k\}$$

In the software context, a component could be a computer program, a reusable program component, a module, a class or object, or even a programming language statement.

System Modeling

System engineering is a modeling process

For each view:

- Define processes
- Represent process behavior
- List process assumptions
- Define external and internal inputs
- Model linkages (control, data, I/O)

System Modeling [2]

Assumptions

- range of allowable data

Simplifications

- partition data into categories

Limitations

- bounds on functionality

Constraints

- guide the implementation

Preferences

- indicate preferred architecture (data, functions, technology)

Much of this information is derived from customer requirements

Business Process Engineering

Data Architecture

- Framework for the data objects used by the business (+ relationships)

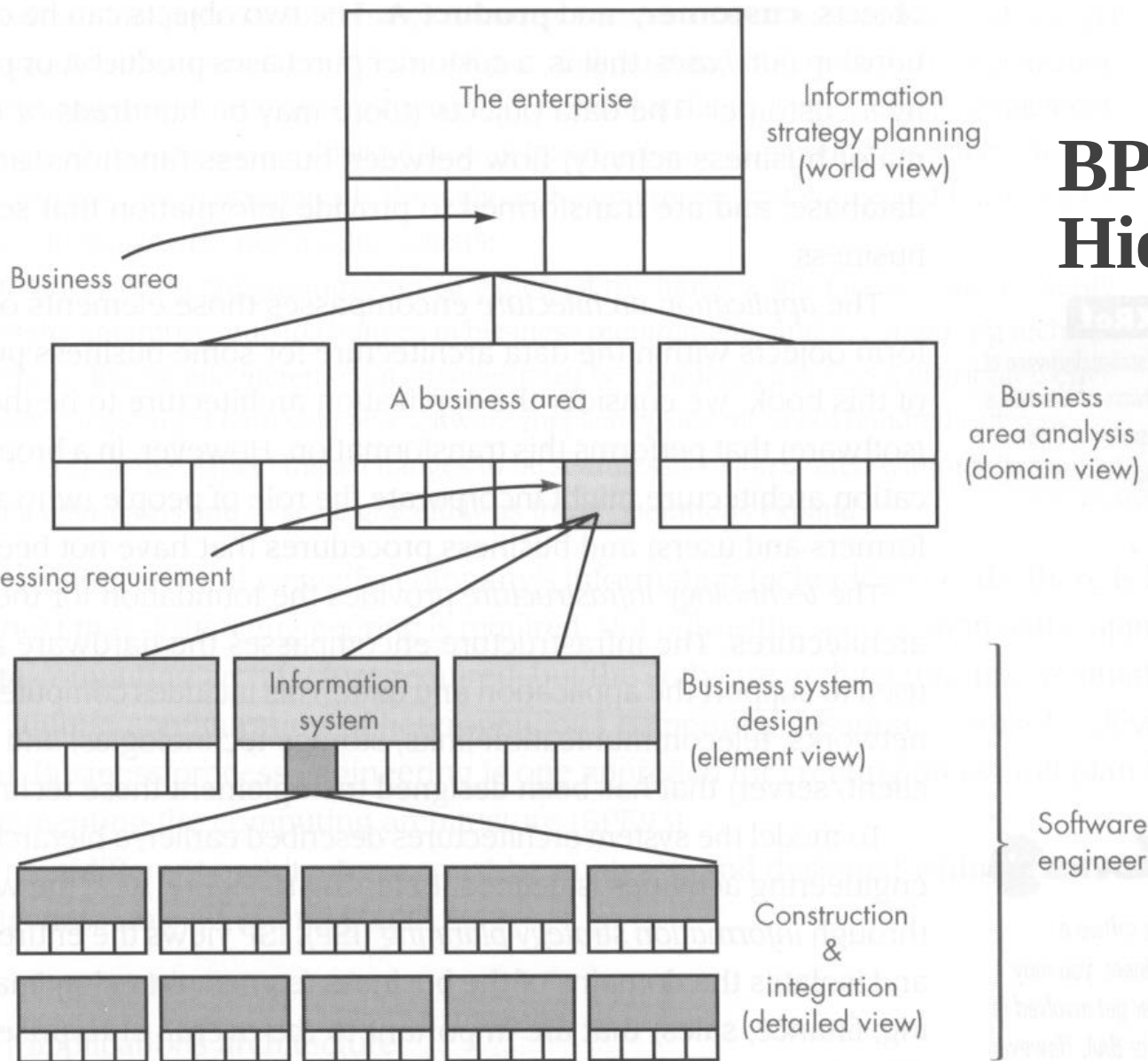
Applications Architecture

- Elements which transform data objects for a business purpose

Technology Infrastructure

- Foundation for data & applications architectures

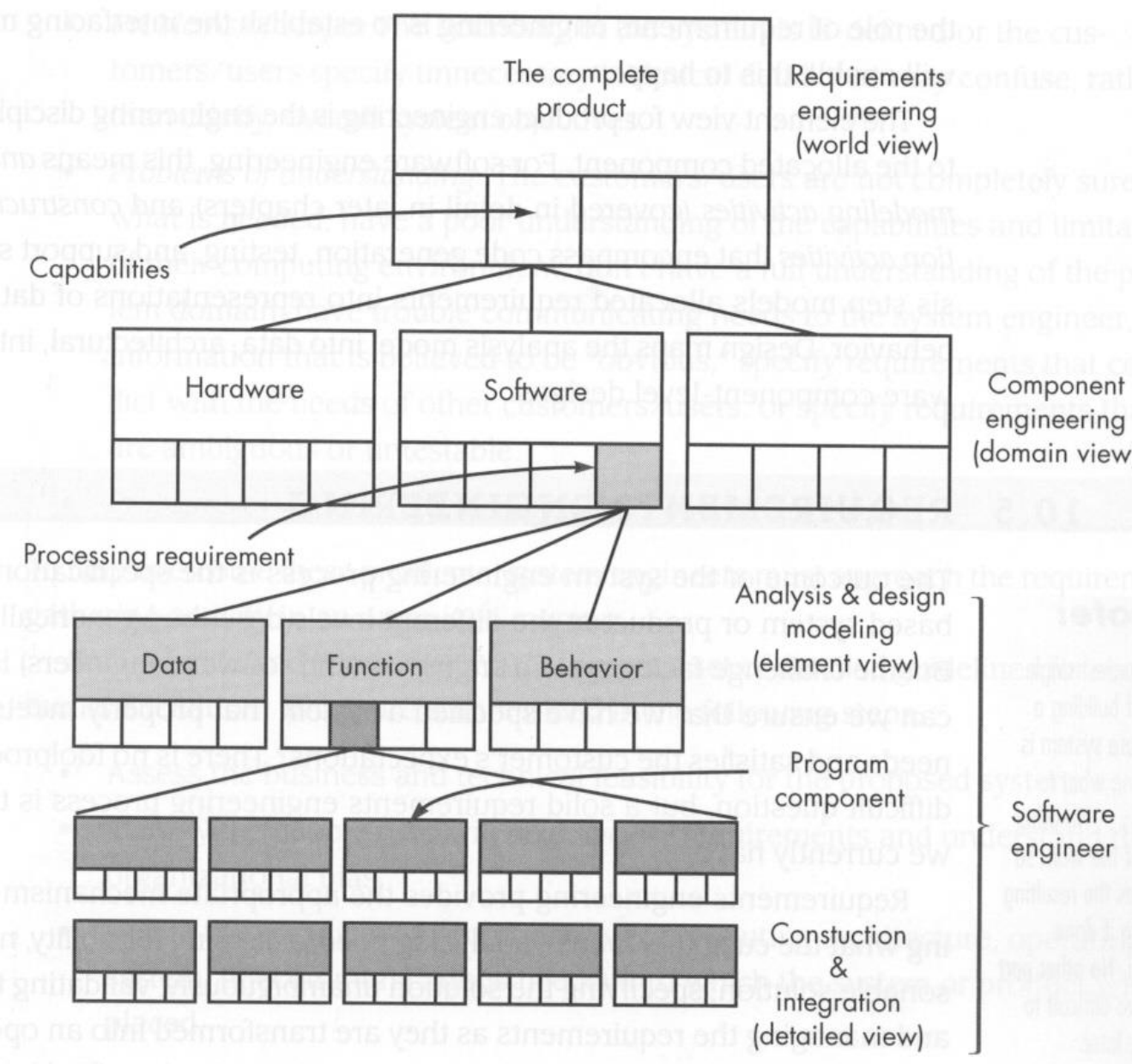
BPE Hierarchy



Product Engineering

The goal of product engineering is to translate the desire for a set of defined capabilities into a working product. To achieve this goal, product engineering—like business process engineering—must derive architecture and infrastructure. The architecture encompasses four distinct system components: software, hardware, data (and databases), and people. A support infrastructure is established and includes the technology required to tie the components together and the information (e.g., documents, CD-ROM, video) that is used to support the components.

Product Engineering Hierarchy



[From SEPA 5/e]

Requirements Engineering

Elicitation

Modeling

Analysis & Negotiation

Validation

Specification

Management

How can we specify a system that meets the customer's needs and expectations?

Requirements Elicitation

Challenges

- Scope:
 - Defining the system boundary
 - Lack of clarity on overall objectives
- Understanding:
 - Customer not skilled
 - Doesn't state the obvious
 - Requirements ambiguous, conflicting, ...
- Volatility:
 - Requirements change over time

Elicitation [2]

Assess feasibility

Identify people & their role(s)

Define technical environment

Identify domain constraints

Select elicitation method(s)

Solicit participation from
several perspectives

Identify ambiguous
requirements

Create usage scenarios

Analysis & Negotiation

Is each requirement:

- Consistent with overall objective?
- Sufficiently abstract?
- Essential to overall objective?
- Bounded and unambiguous?
- Attributed to a source? (person)
- Conflicting with other requirements?
- Achievable in technical environment?
- Testable, once implemented?

Requirements Specification

Elements of a Specification:

- Written documents
- Graphical models
- Formal mathematical models

Final work product:

System Specification

System Modeling

Evaluate the system's components in relation to one another

Link requirements to system components

Validate assumptions about data flow, work flow, input / output, ...

Requirements Validation

Is each requirement:

- Stated clearly?
- Verified by an identified source?
- Bounded in a quantitative way?
- Associated with other requirements?
- Consistent with domain constraints?
- Testable, with specified tests?
- Traceable to the system model?
- Traceable to overall objectives?

Requirements Management

Identify, control, and track:

- New requirements
- Changes to requirements

Active throughout the life-cycle

Traceability Table

- Relates requirements to features, source, dependency, subsystem, interface, etc.

Thank You